Developing Species Distribution Models in R

Guest Lecture: Kesse Asante

What is species distribution modeling (SDM)?

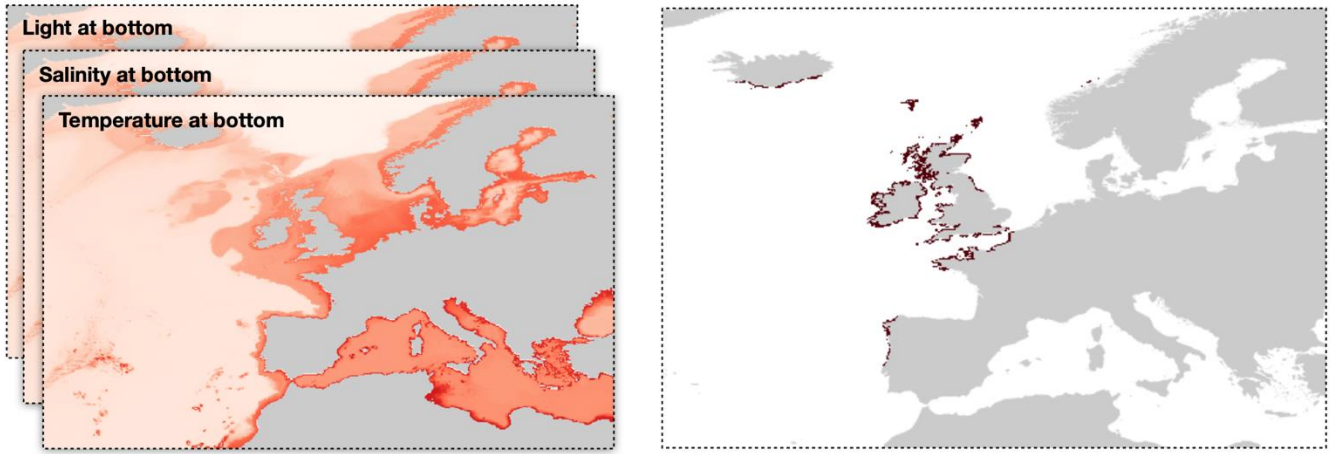
Process of **using computer algorithms to estimate and predict the relationship between species distributions** (occurrence, abundance) **and the environmental**.

Purpose:

- Gain insights into species' environmental tolerances and habitat preferences.
- Make spatial predictions of geographic distributions across space and time.

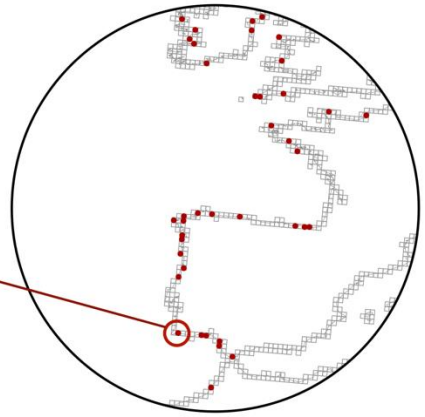
Assis (2024)

Species Records + environmental layers

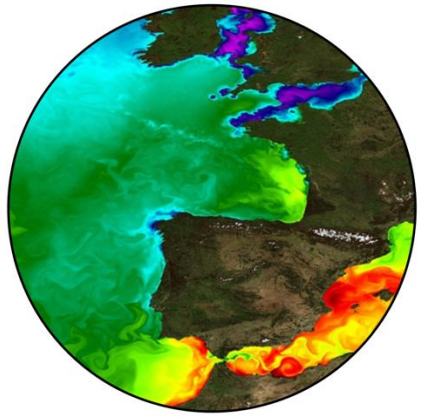


The observed distribution reflects species' environmental requirements.

	SST	Salinity	Oxygen
Site 1	21.0	35.4	210.1
Site 2	19.4	29.1	224.9
Site 3	21.5	31.1	199.8
Site 4	21.3	33.4	201.1
Site 5	14.8	34.1	184.8
(...)	(...)	(...)	(...)



Records of species distribution



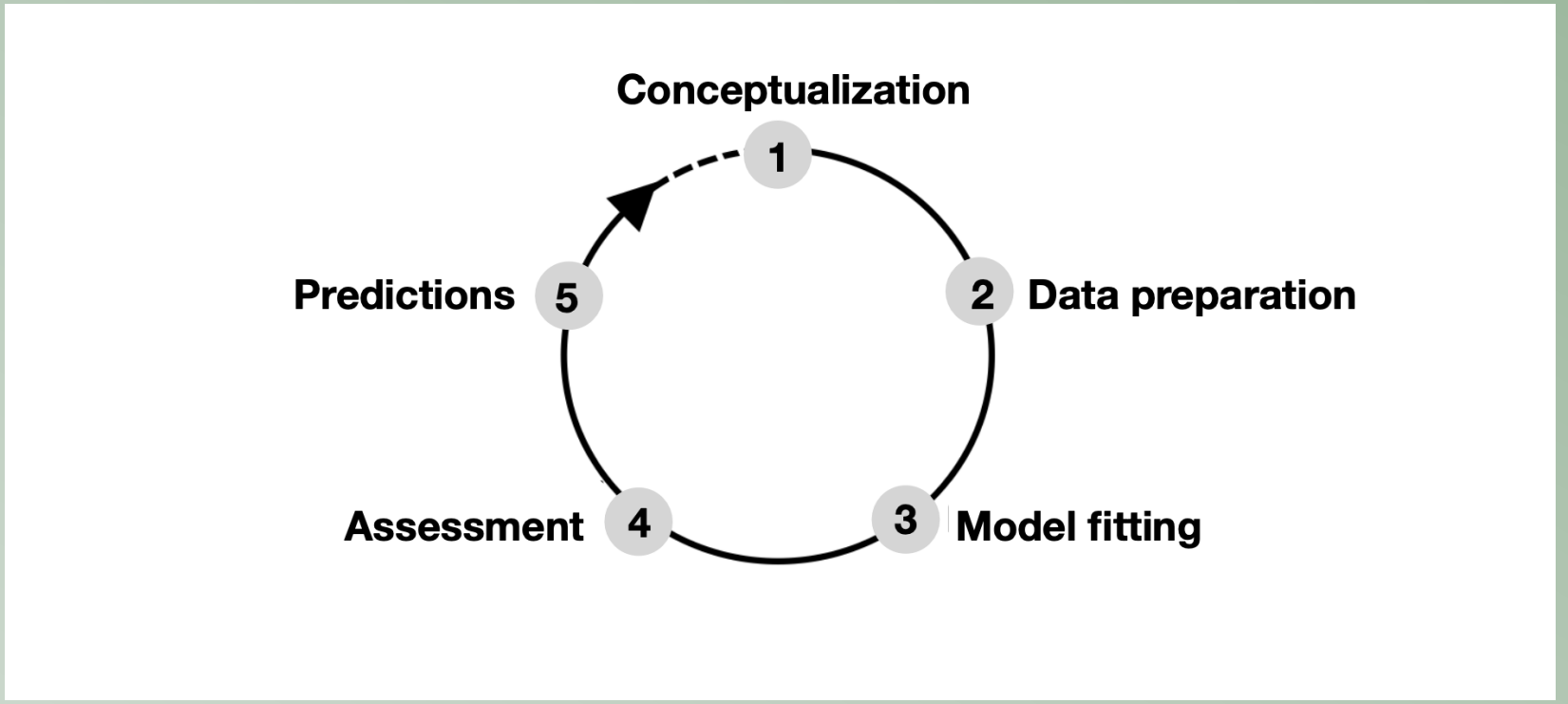
Environmental data

Using environmental conditions to explain species distribution.

Why SDMs matter (marine focus)

- **Increasingly vital tools** in marine science and resource management.
- **Ecology**: understand the drivers and patterns of species distributions.
- **Conservation planning and Marine Spatial Planning**: identify areas for conservation (e.g.,
- MPA design); evaluate the potential of invasive species to settle in new areas; identify suitable areas for aquaculture.
- Fisheries management: identify suitable areas of commercial species (e.g., habitat mapping).
- Climate change ecology: project climate change impacts (e.g., feed IPCC reports).

Model Building



The goal is to build ecologically meaningful models fit for the purpose (e.g., conservation planning and climate impact assessment).

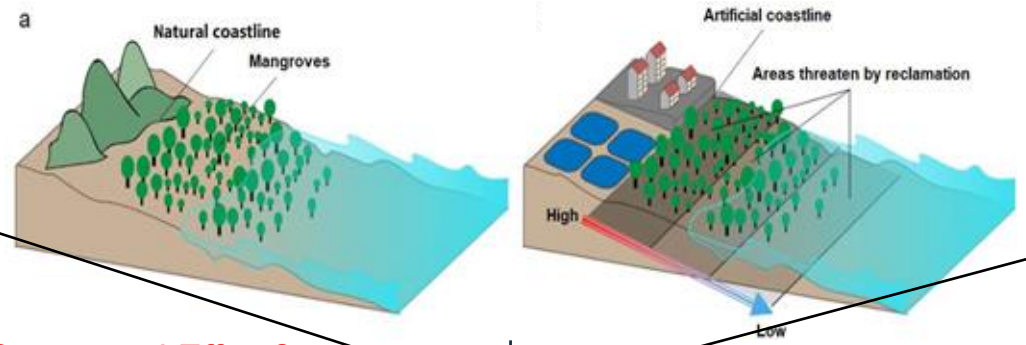


Climate Change and SAI

Change?

Change?

Coastal Landscape?



Sea level Variations?



Impacts on coastal communities and low-lying islands?

Air Temperature and Precipitation?

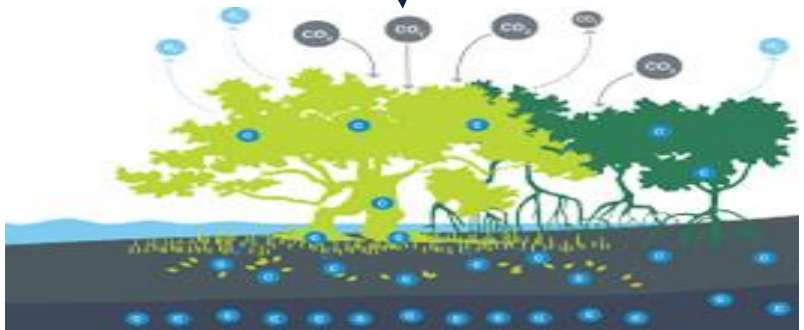


Global Mangrove Status: Projected Changes of Extent, Distribution, and Biomass?

Compound Effect?



Fate of Blue Carbon, Mangrove Carbon Stocks?



Environmental variables: selection

Ecological relevance is key:

- Choose variables known or hypothesized to directly influence the species' survival,
- growth, or reproduction.
- Consider physiology and life history.
- Using large datasets is an approach with stronger criticism.

Important marine variables:

- Physical: temperature, salinity, bathymetry, currents velocity, sea ice conditions.
- Chemical: oxygen, nutrients (e.g., nitrate and phosphate).
- Biological: chlorophyll-a (proxy for productivity).

LAB

Building an SDM for Seagrass and Corals

Model fitting

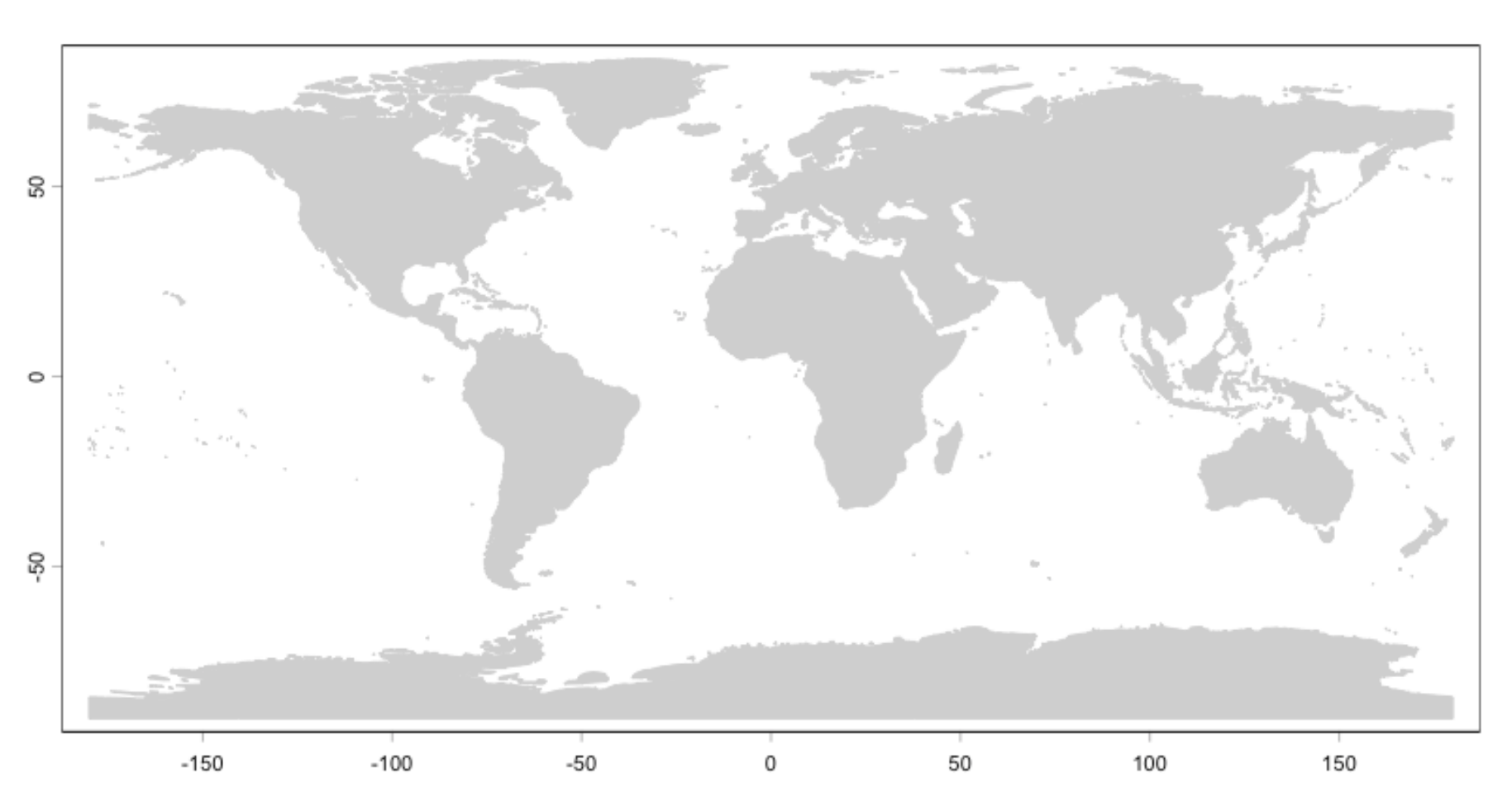
Model fitting (i.e., the **relationship between species occurrence and environmental predictors**) is the core of SDM.

Presence/ absence data

```
# Install packages from CRAN
install.packages(c("terra", "rgbif", "robis", "RJSONIO", "ecospat", "blockCV", "xgboost","devtools"))
# Install package from GitHub
devtools::install_github("bio-oracle/biooracler")
#install SDM Package
install.packages("/Users/new/Desktop/Sci_Comp26/bDSSDMTools_0.0.0.9000.tar.gz",  repos = NULL, type = "source")

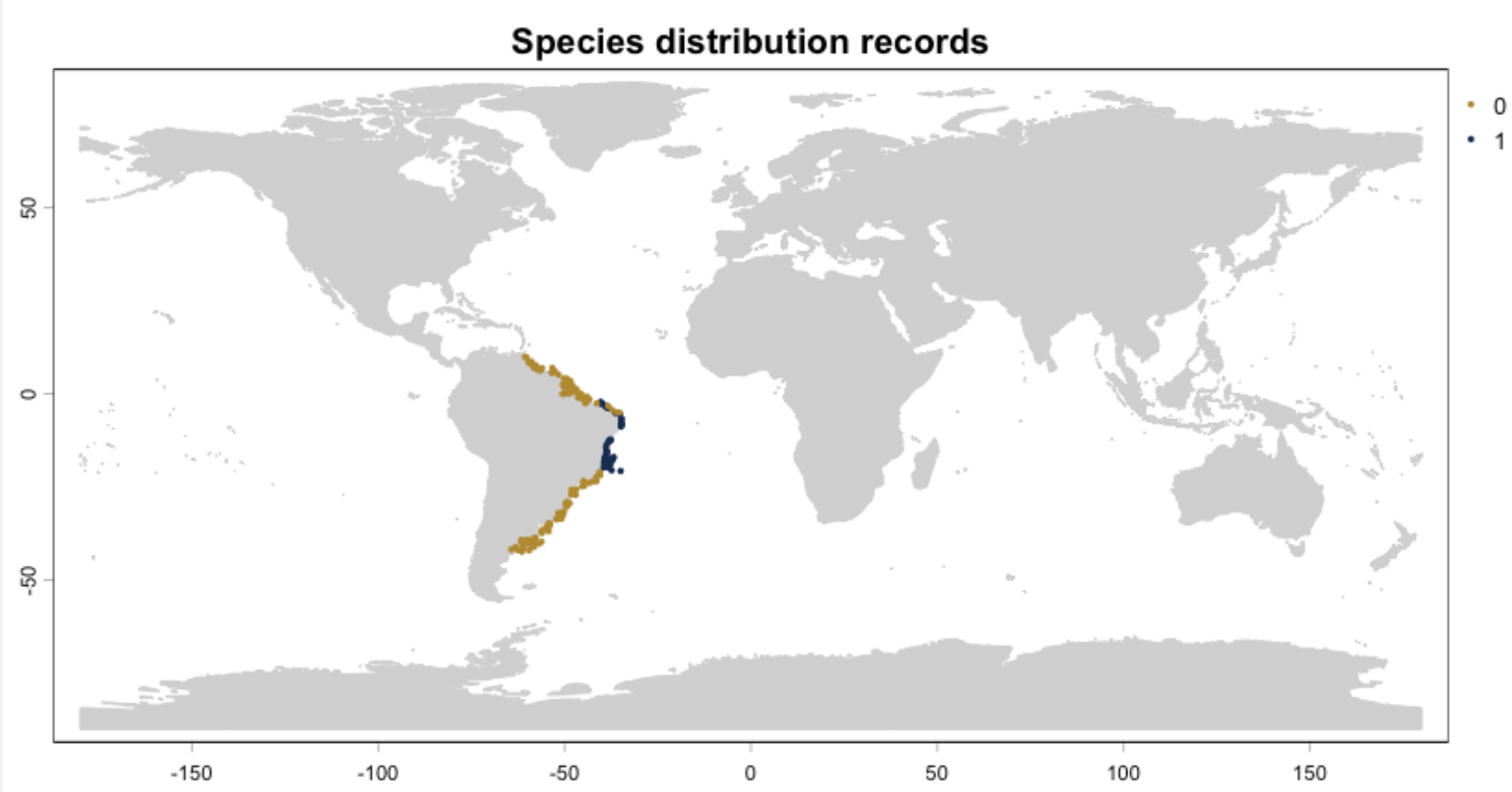
#Load packages
library(terra)
library(bDSSDMTools)
library(ggplot2)
library(sf)
```

```r
## Seagrass
# load a vector defining global landmasses
landmass <- vect("/Users/new/Desktop/Sci_Comp26/Data/Landmass/Landmass.shp")
# plot landmass
plot(landmass, col = "#D4D4D4", border = "#D4D4D4", axes = TRUE) #hex color codes; col = color and border = border line color
```



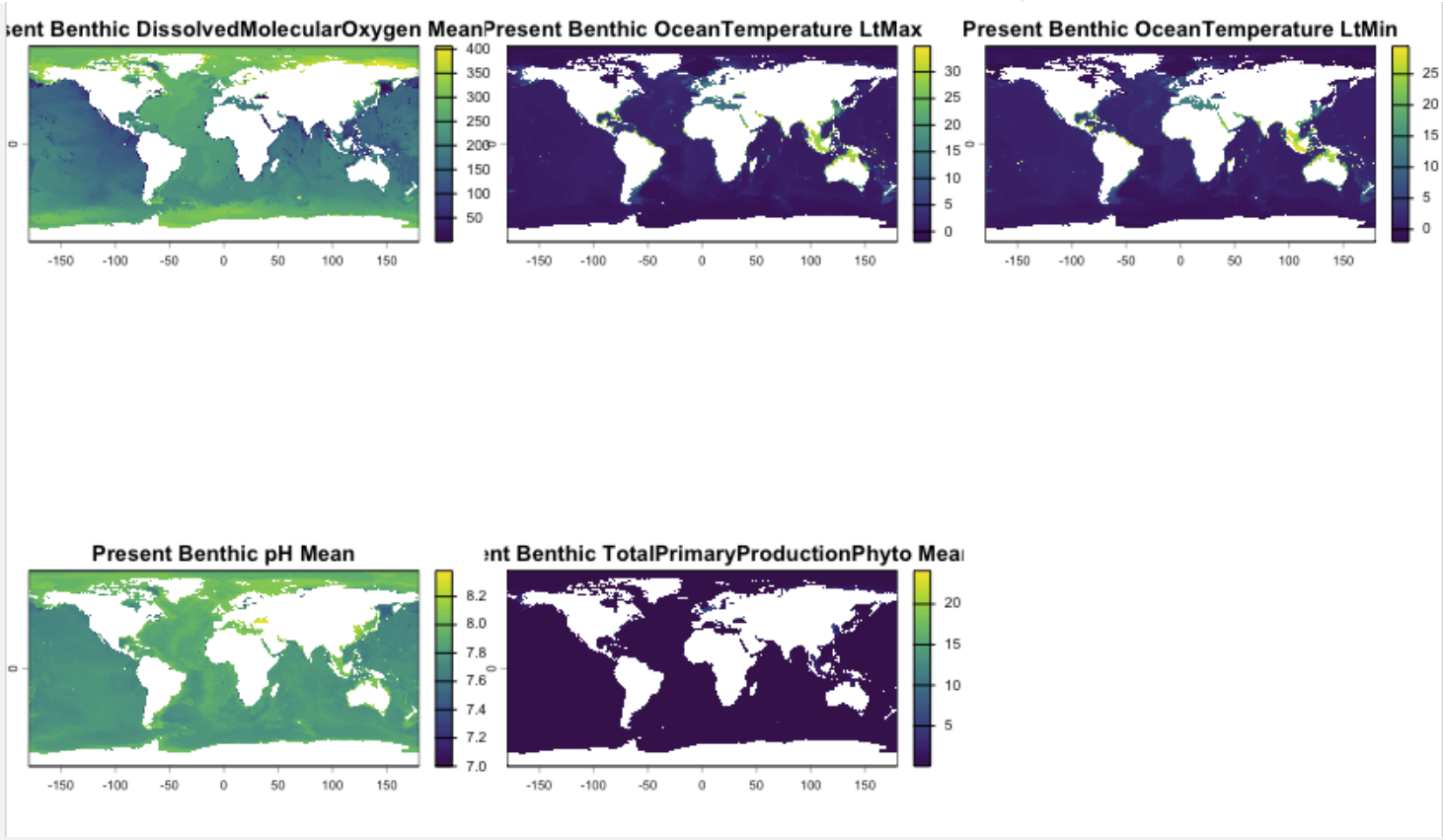
```r
#Seagrass Records
#Let's take peak at how the records file looks like and how elements are separated. CSV default separator is ","
readLines("/Users/new/Desktop/Sci_Comp26/Data/Seagrass/species_presence_absence_records.csv", n = 3)
records <- read.csv("/Users/new/Desktop/Sci_Comp26/Data/Seagrass/species_presence_absence_records.csv", sep = ";")
#transform data.frame into a spatial object
records <- vect(records, geom = c("Lon", "Lat"))

#plot record
plot(landmass, main = "Species distribution records", col = "#D4D4D4", border = "#D4D4D4", axes = TRUE)
plot(records, y = "PA", col = c("#c29431", "#043259"), pch = 16, cex = 0.5, add = TRUE) #pch = plotting character & cex = size
```

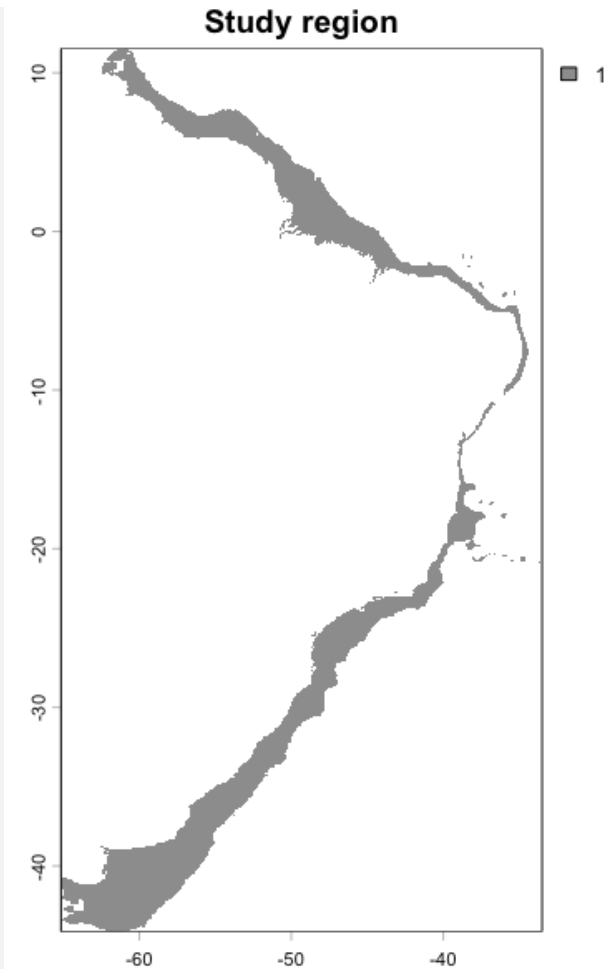


```r
# import environmental data
files <- c("/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic DissolvedMolecularOxygen Mean.tif", # Oxygen
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic OceanTemperature LtMax.tif", # Temperature Max
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic OceanTemperature LtMin.tif", # Temperature Min
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic pH Mean.tif", # pH
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic TotalPrimaryProductionPhyto Mean.tif" # Productivity
)

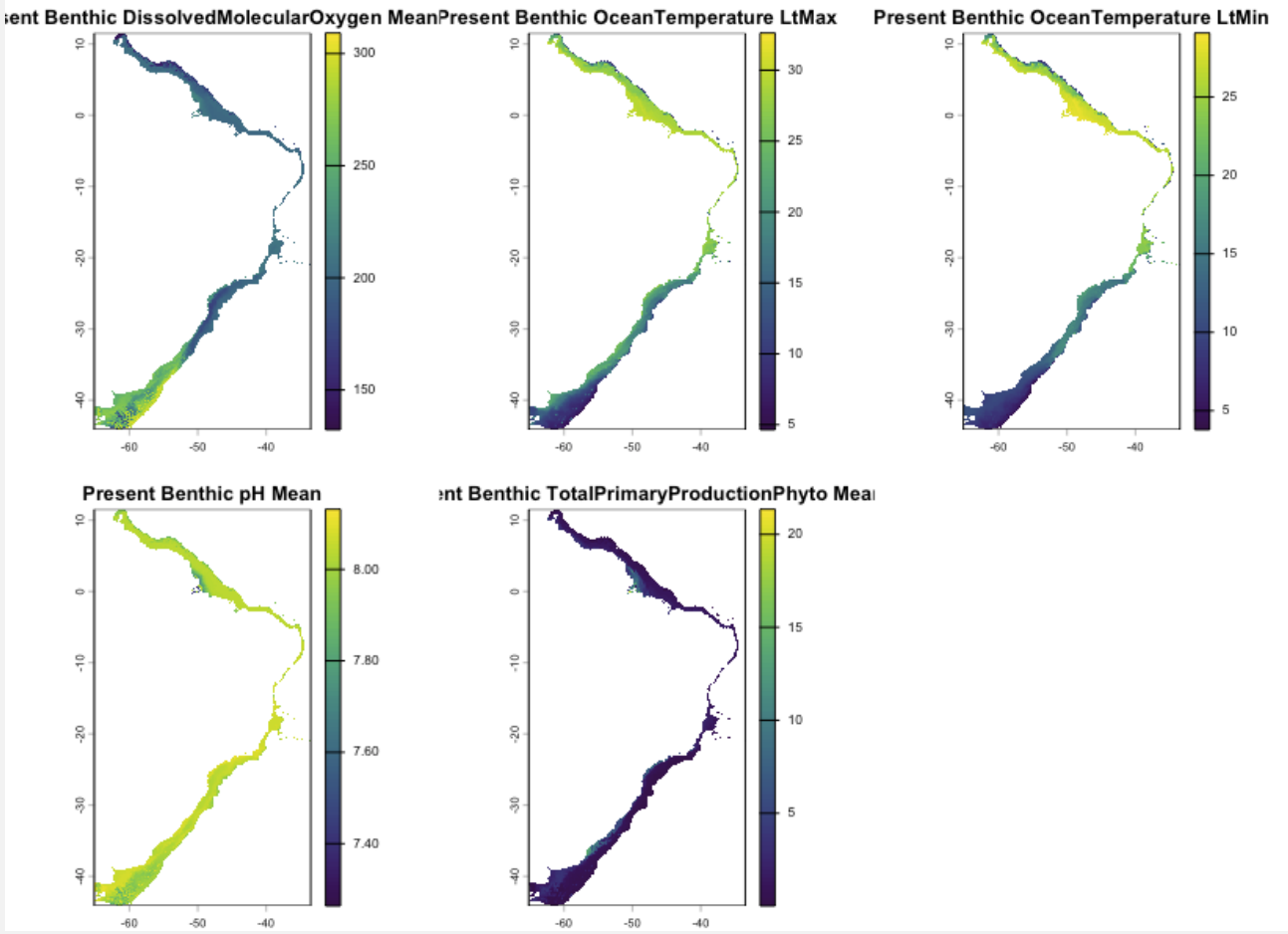
# produce and plot raster layers
environmentalLayers <- rast(files)
plot(environmentalLayers)
```



```r
# develop a study region layer
myStudyRegion <- studyRegion(rasterLayers=environmentalLayers,
                             presences=records, # presence records
                             distanceThreshold=200000, # distance from records in meters
                             bathymetryLayer = "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Bathymetry.tif", # bathymetry layer
                             depthPref=c(0,-500), # known vertical distribution
                             intertidalLayer = "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/CoastLine.tif" ) # consider intertidal
# inspect the study region layer
plot(myStudyRegion, main="Study region", col="#979797", axes=TRUE)
```



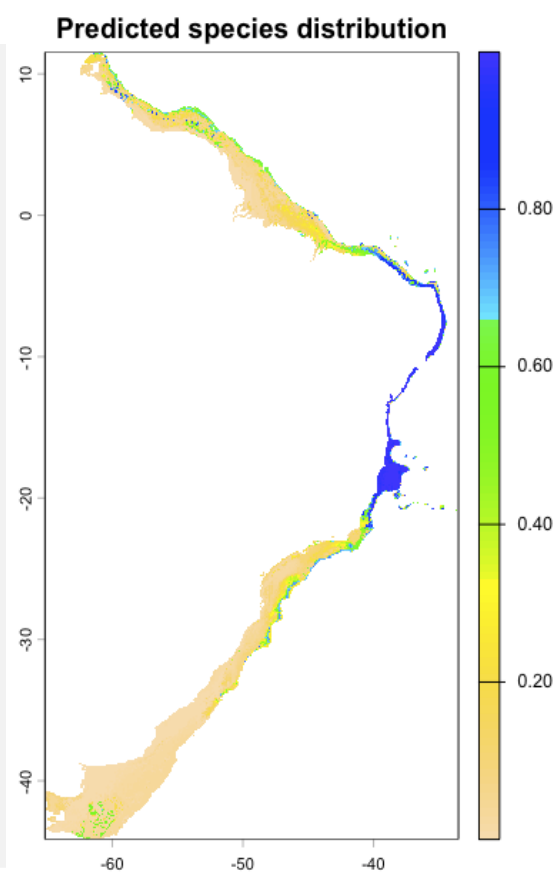
```r
# crop the environmental layers to the extent of the study region
environmentalLayers <- crop(environmentalLayers, myStudyRegion)
# mask the environmental layers with the study region
environmentalLayers <- mask(environmentalLayers, myStudyRegion)
# plot the masked environmental layers
plot(environmentalLayers, axes = TRUE)
```

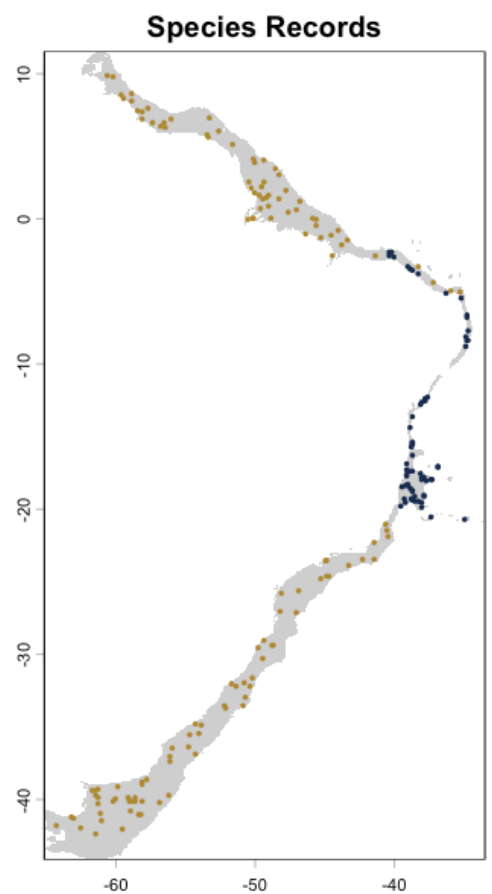


```r
# extract environmental values and make an object with all information to model
modelData <- generateModelData(records, envConditions = environmentalLayers)

# fit a binomial GLM to the dataset
model <- trainModel(modelData, algorithm = "brt")
# predict the distribution with the model over a raster stack
prediction <- predictModel(model = model, newData = environmentalLayers)

# map the predicted distribution
plot(prediction$rasterLayer, main = "Predicted species distribution", col = rev(topo.colors(100)))
```

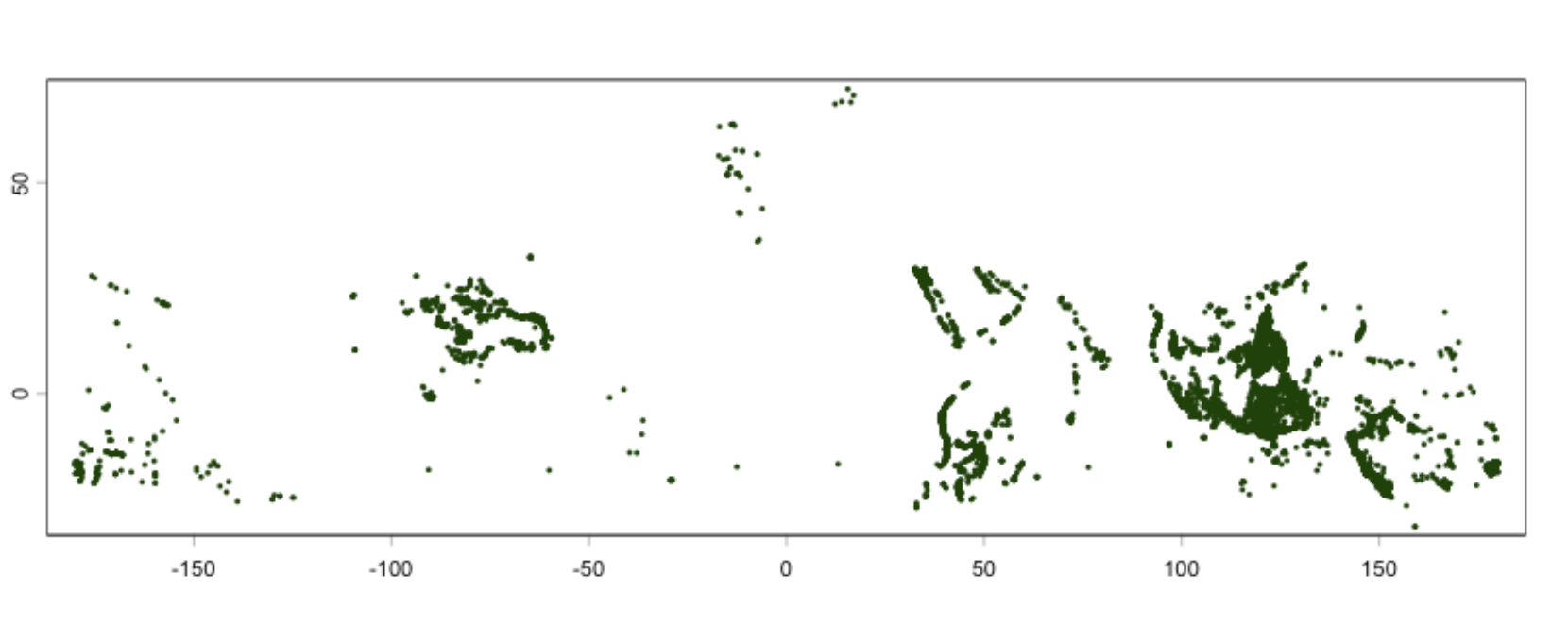




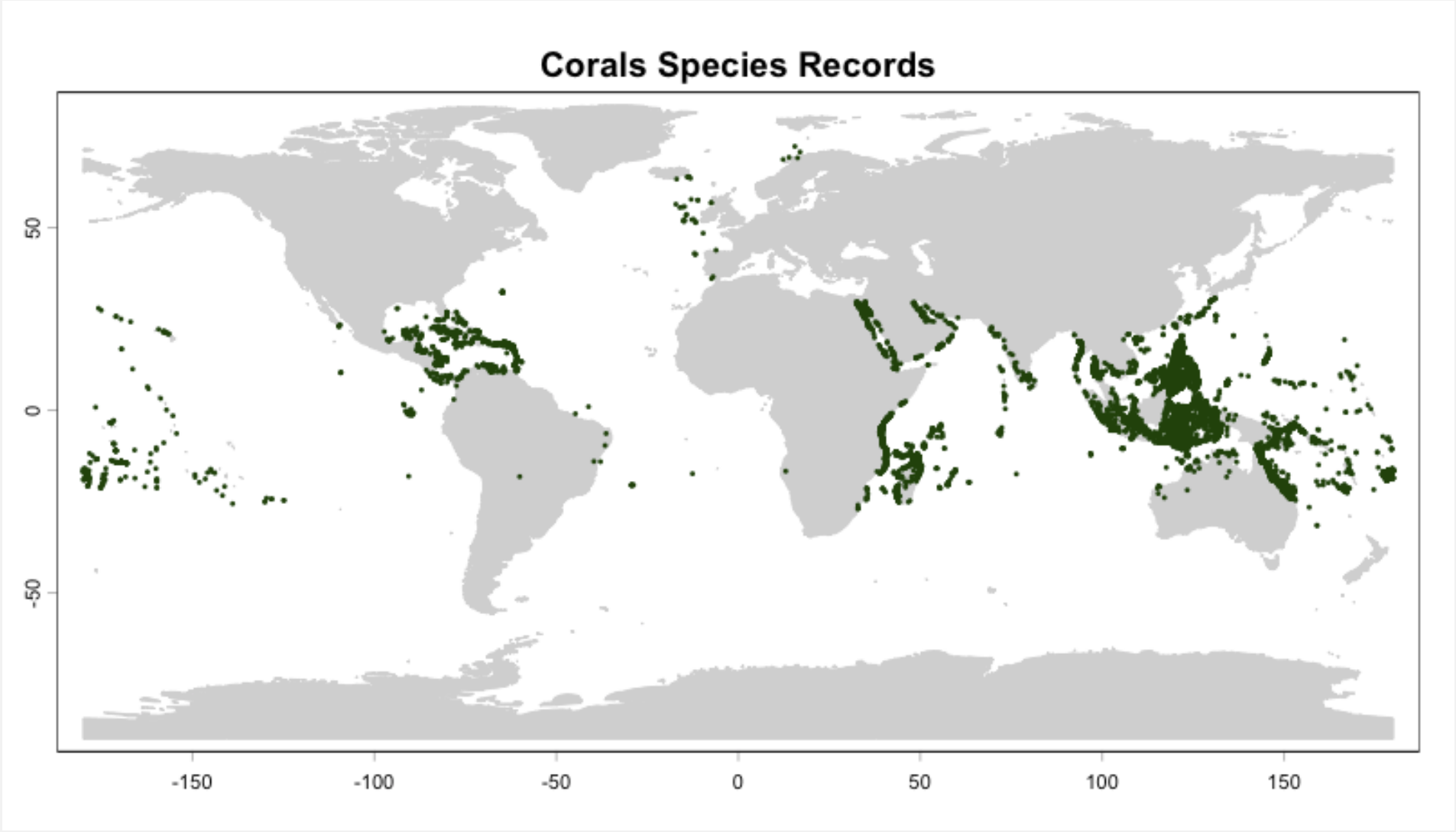
Presence / pseudo-absence data

If **true absence data is not available**, but one wants to fit a brt model, **pseudo-absence** data can be generated.

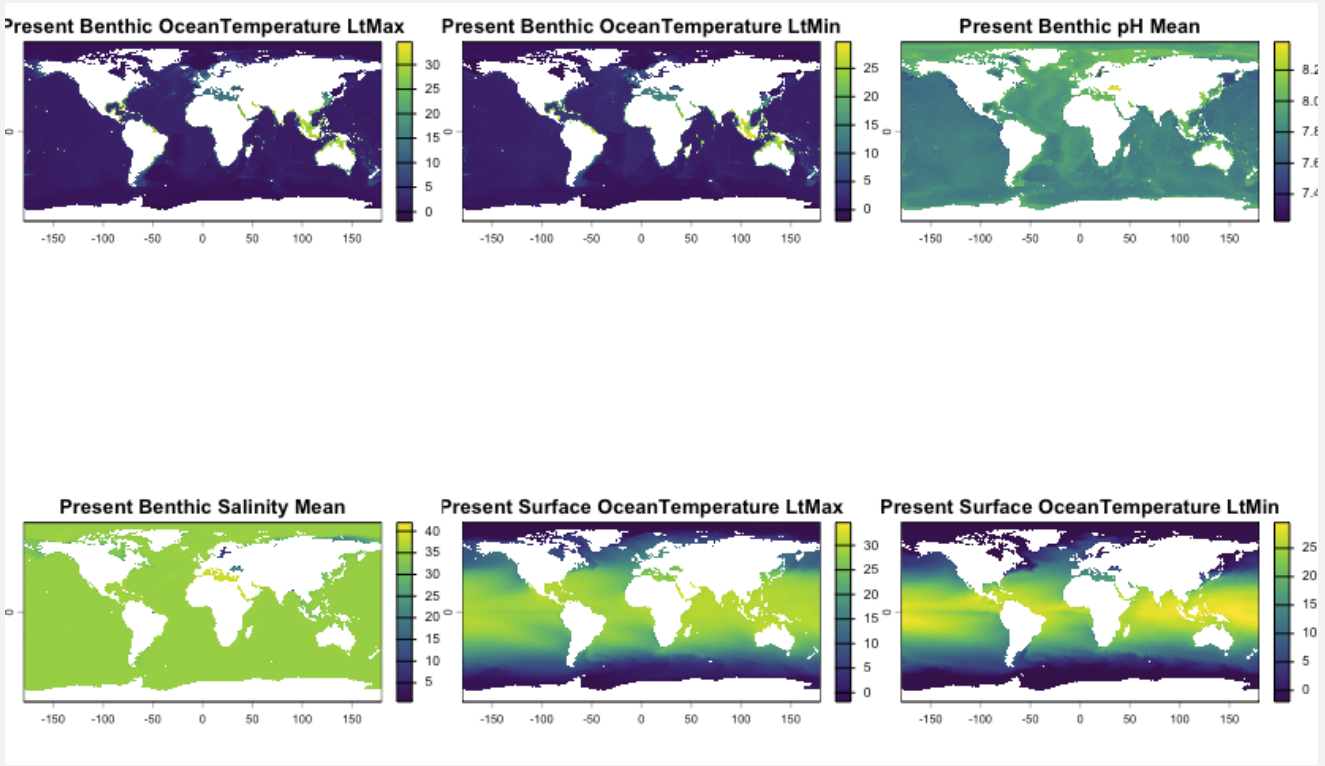
```
#Corals occurrence (Source: UNEP-WCMC)
presences <- read.csv("/Users/new/Desktop/Sci_Comp26/Data/Corals/occurrenceRecords.csv")
#transform data.frame into a spatial object
presences <- vect(presences, geom = c("Lon", "Lat"))
#plot occurrences
plot(presences, col = "#034B03", pch = 16, cex = 0.4) #pch = plotting character & cex = size
```



```r
#Plot records on landmass
plot(landmass, main = "Corals Species Records", col = "#D4D4D4", border = "#D4D4D4", axes = TRUE)
plot(presences, col = "#034B03", pch = 16, cex = 0.4, add = TRUE)
```

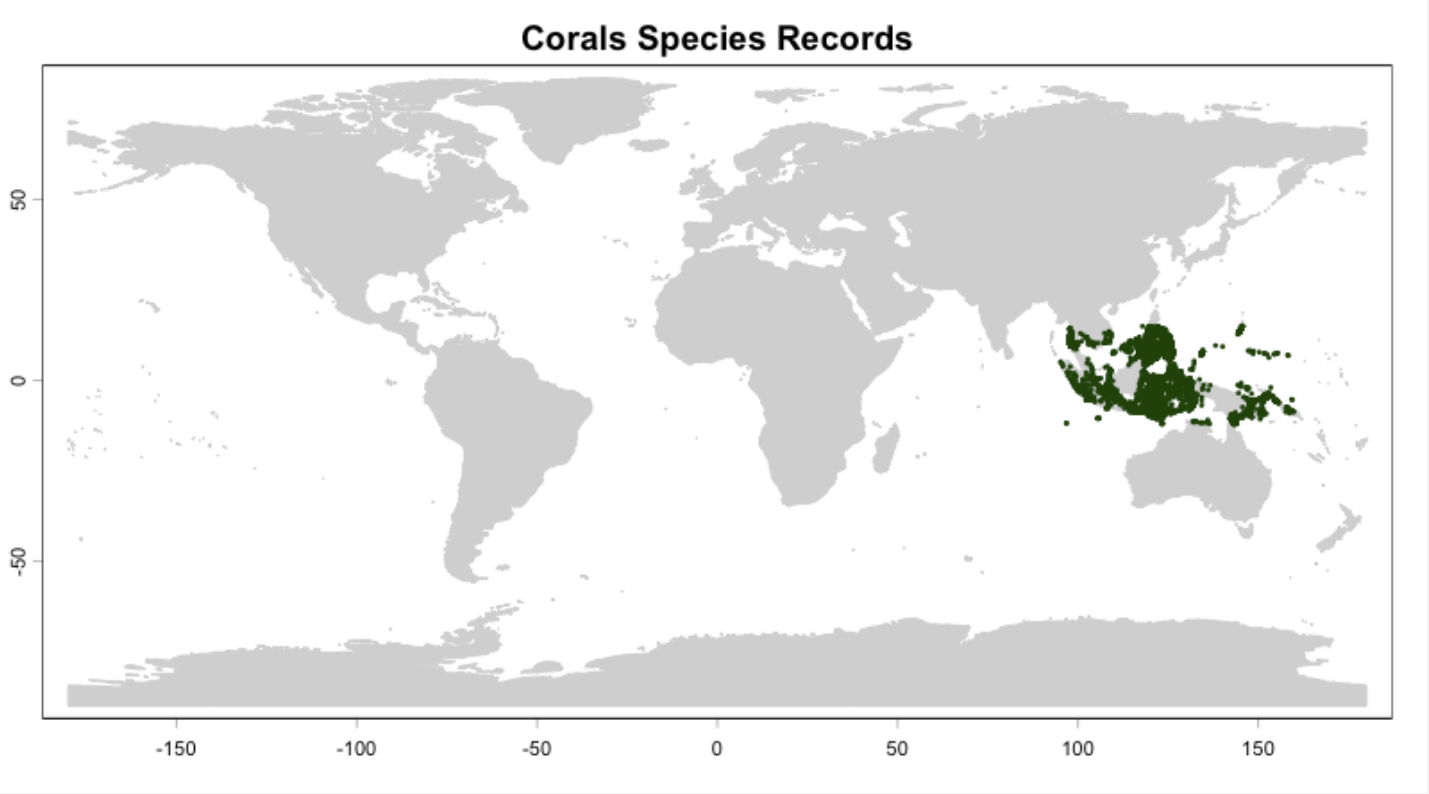


```r
# import environmental data
files <- c("/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic OceanTemperature LtMax.tif",
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic OceanTemperature LtMin.tif",
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic pH Mean.tif",
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Benthic Salinity Mean.tif",
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Surface OceanTemperature LtMax.tif",
           "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Present Surface OceanTemperature LtMin.tif"
)
# produce and plot raster layers
environmentalLayers <- rast(files)
plot(environmentalLayers)
```



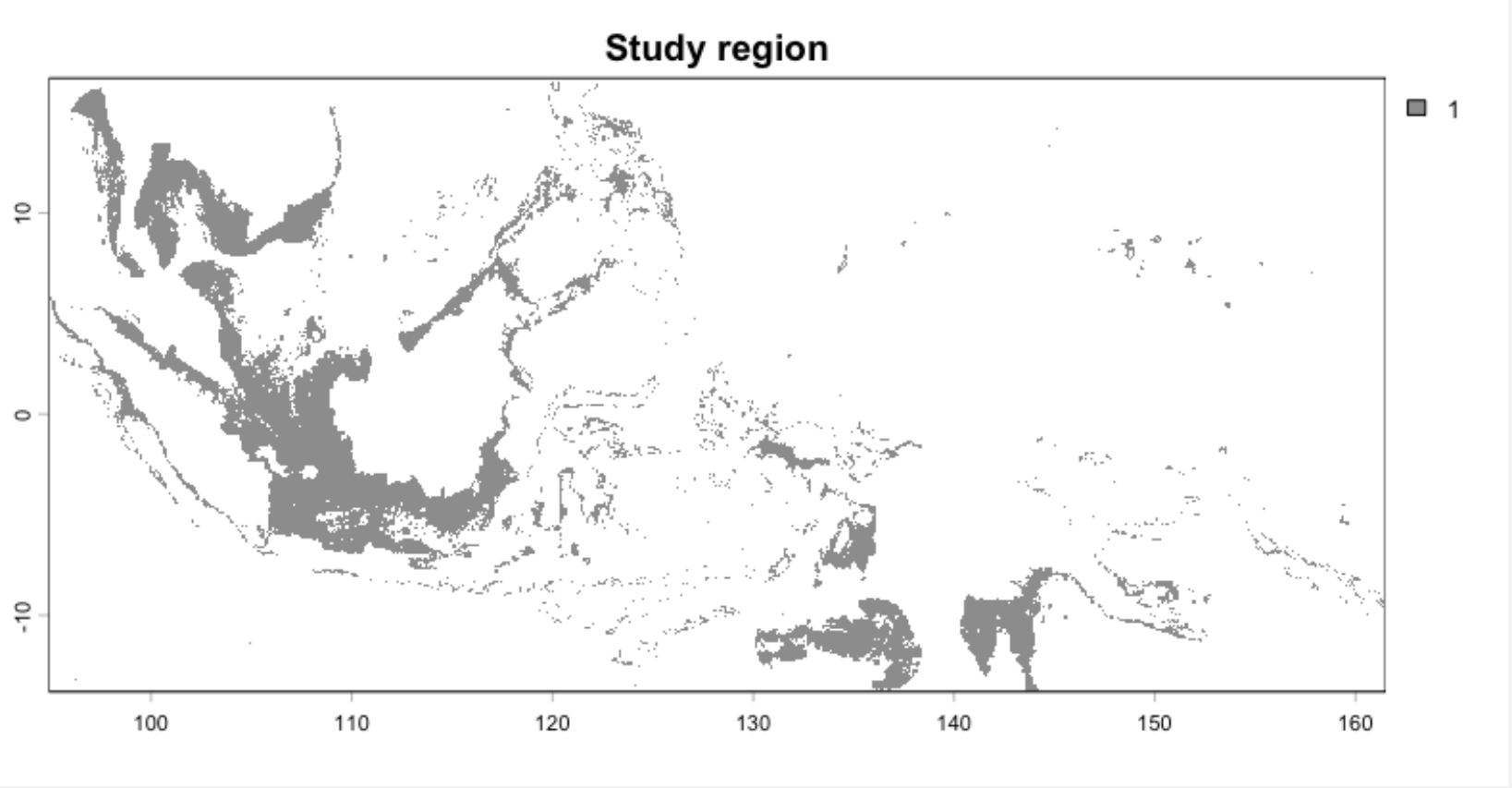
```r
#Defining the extent as study area
ct_extent <- ext(95, 160, -12, 15)   # xmin, xmax, ymin, ymax
# Crop presence points to IWP
presences_ct <- crop(presences, ct_extent)
plot(presences_ct)

#Look at where it is on the landmass
plot(landmass, main = "Corals Species Records", col = "#D4D4D4", border = "#D4D4D4", axes = TRUE)
plot(presences_ct, col = "#034B03", pch = 16, cex = 0.4, add = TRUE)
```

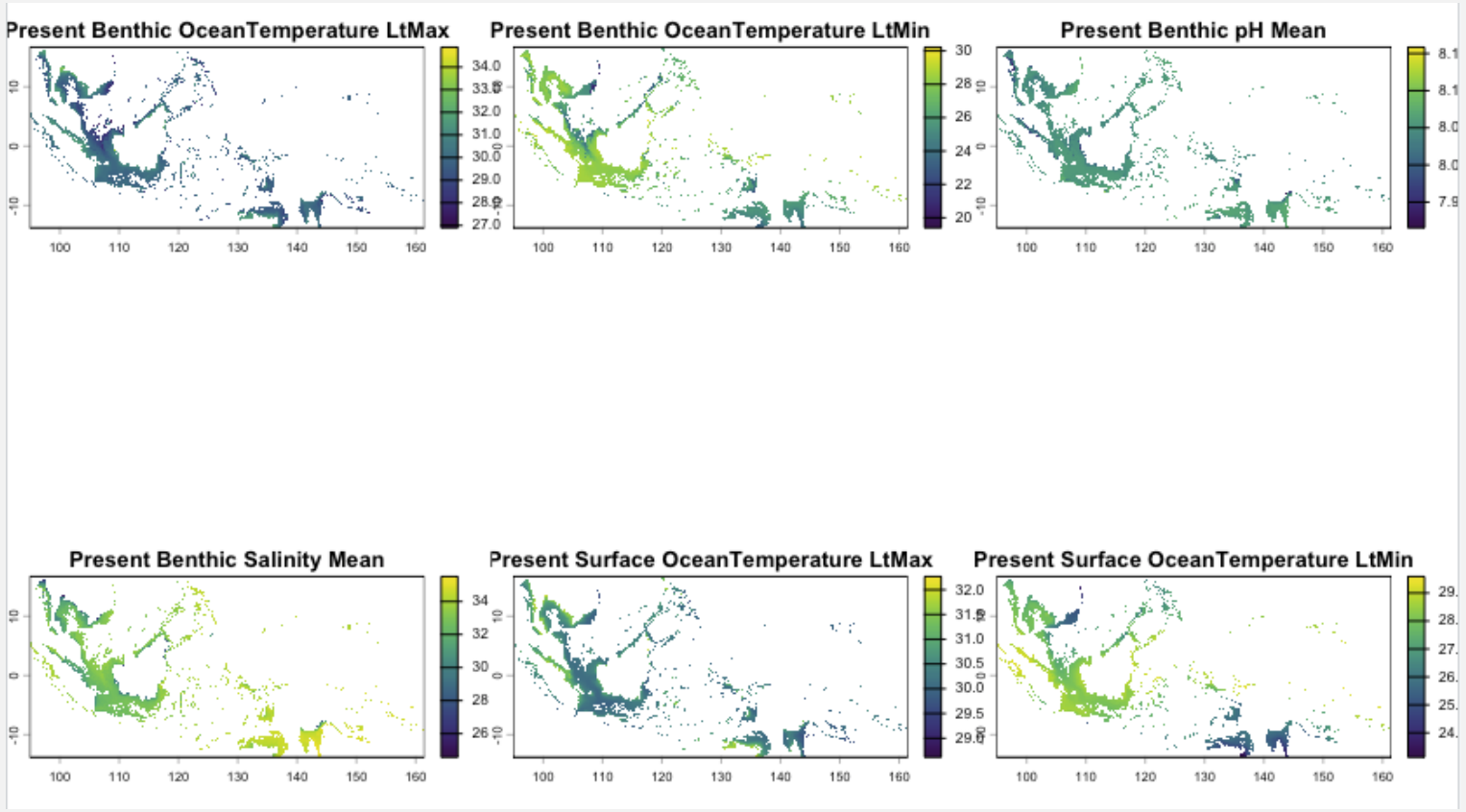


```r
#Develop study region area
myStudyRegion <- studyRegion(rasterLayers = environmentalLayers,
                             presences = presences_ct,
                             distanceThreshold = 200000, #distance from records in meters
                             bathymetryLayer = "/Users/new/Desktop/Sci_Comp26/Data/Raster_data/Bathymetry.tif",
                             depthPref = c(0, -60)) # To cover depth: 0 to 40 m for reef-building corals

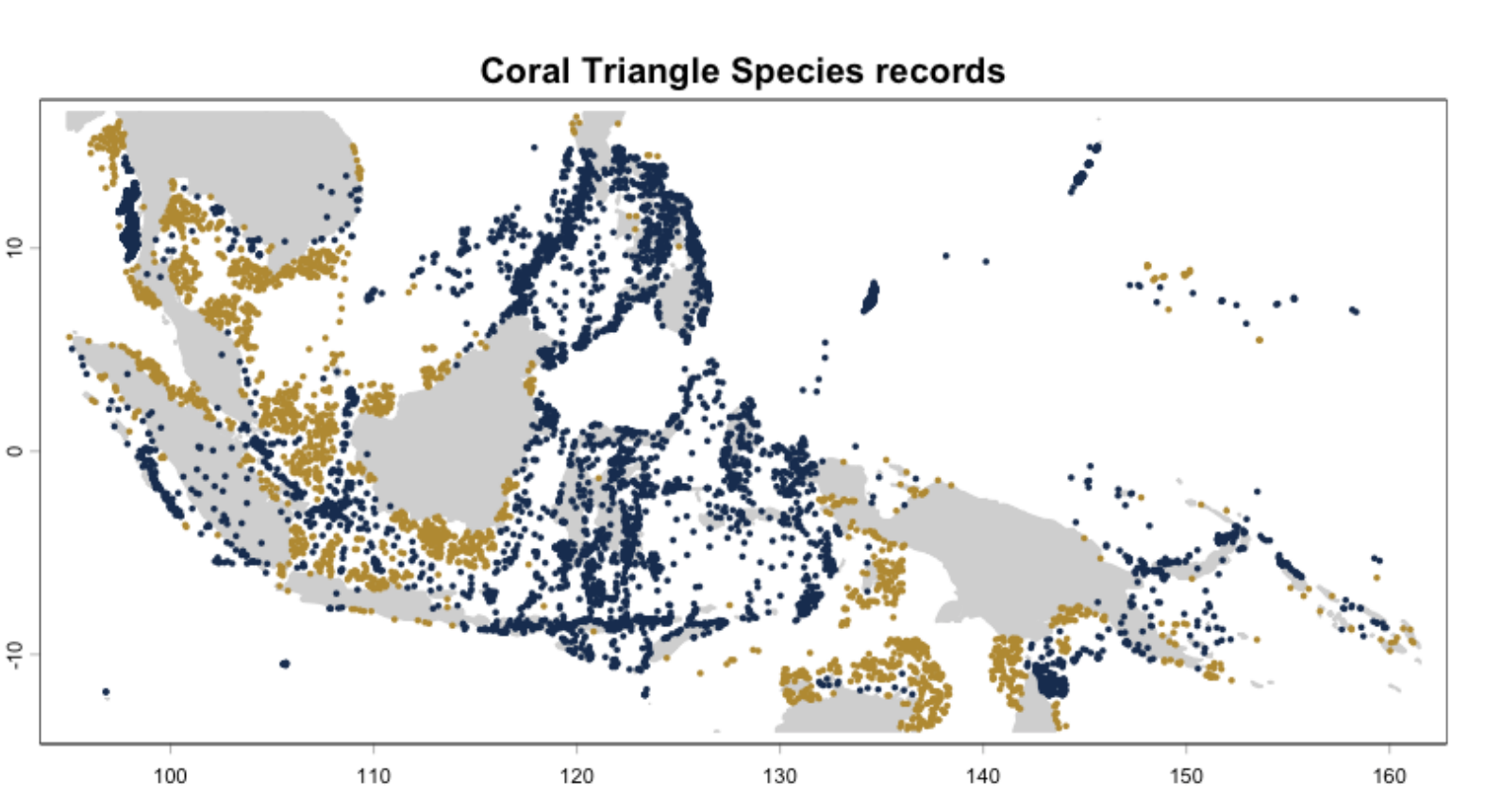
# inspect the study region layer
plot(myStudyRegion, main="Study region", col="#979797", axes=TRUE)
```



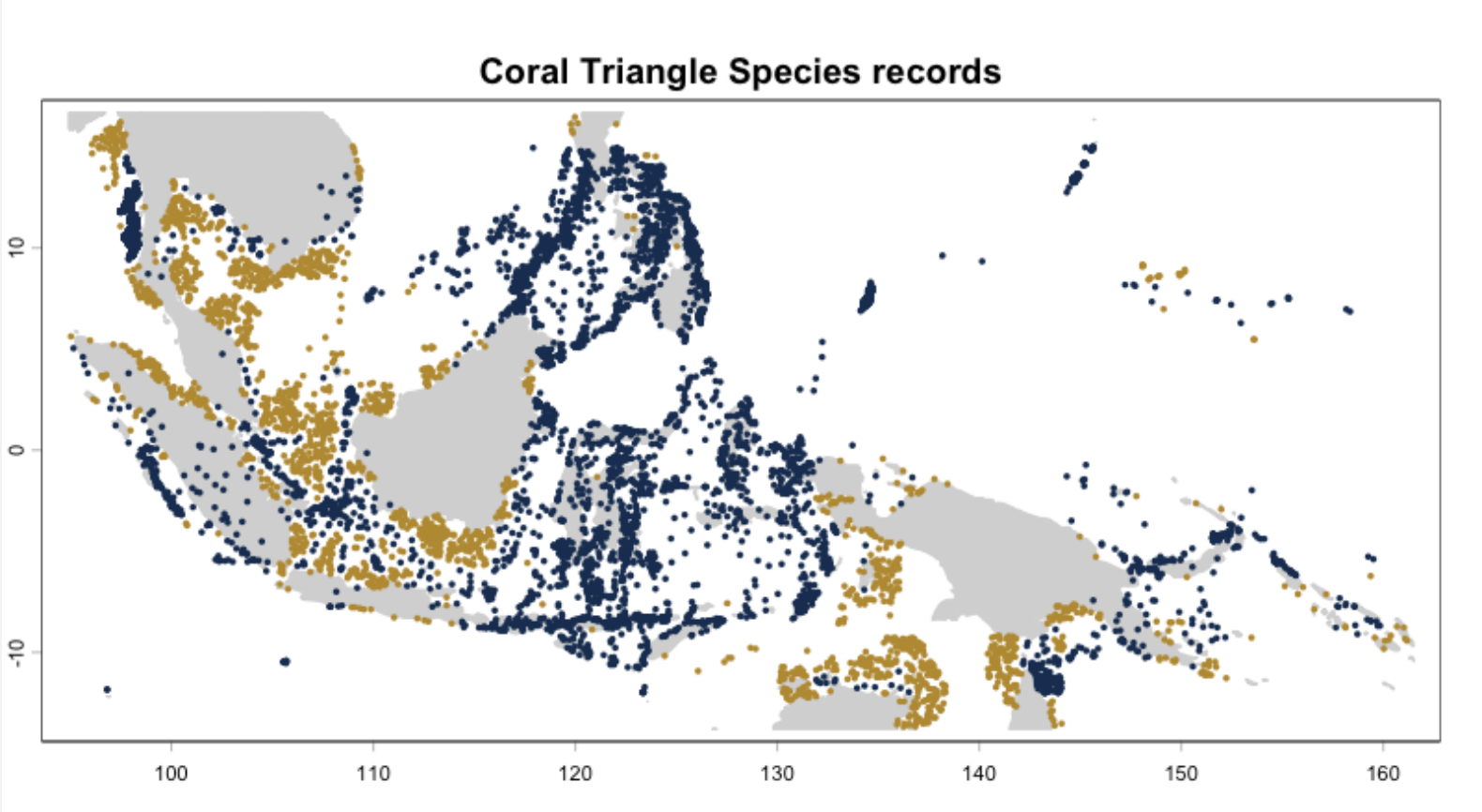
```r
# crop the environmental layers to the extent of the study region
environmentalLayers <- crop(environmentalLayers, myStudyRegion)
# mask the environmental layers with the study region
environmentalLayers <- mask(environmentalLayers, myStudyRegion)
# plot the masked environmental layers
plot(environmentalLayers, axes = TRUE)
```



```r
# generate pseudo-absences
absences <- pseudoAbsences(rasterLayers = environmentalLayers, presences = presences_ct, n = 2000)
# crop landmass to the region for better representation
landmassCT <- crop(landmass, myStudyRegion)
# plot records
plot(landmassCT, main = "Coral Triangle Species records", col = "#D4D4D4", border = "#D4D4D4", axes = TRUE)
plot(absences, col = "#c29431", pch = 16, cex = 0.5, add = TRUE)
plot(presences_ct, col = "#043259", pch = 16, cex = 0.5, add = TRUE)
```



```r
# combine presences and absences into a unique object for modelling
presences_ct$PA <- 1
absences$PA <- 0
records <- rbind(presences_ct, absences)
# plot records
plot(landmassCT, main = "Coral Reefs Species records", col = "#D4D4D4", border = "#D4D4D4", axes = TRUE)
plot(records, y = "PA", col = c("#c29431", "#043259"), pch = 16, cex = 0.5, add = TRUE)
```



```r
# extract environmental values and make an object with all information to model: Model evaluation with cross validation
modelData <- generateModelData(records, envConditions = environmentalLayers, method = "blocks", kFolds = 10)

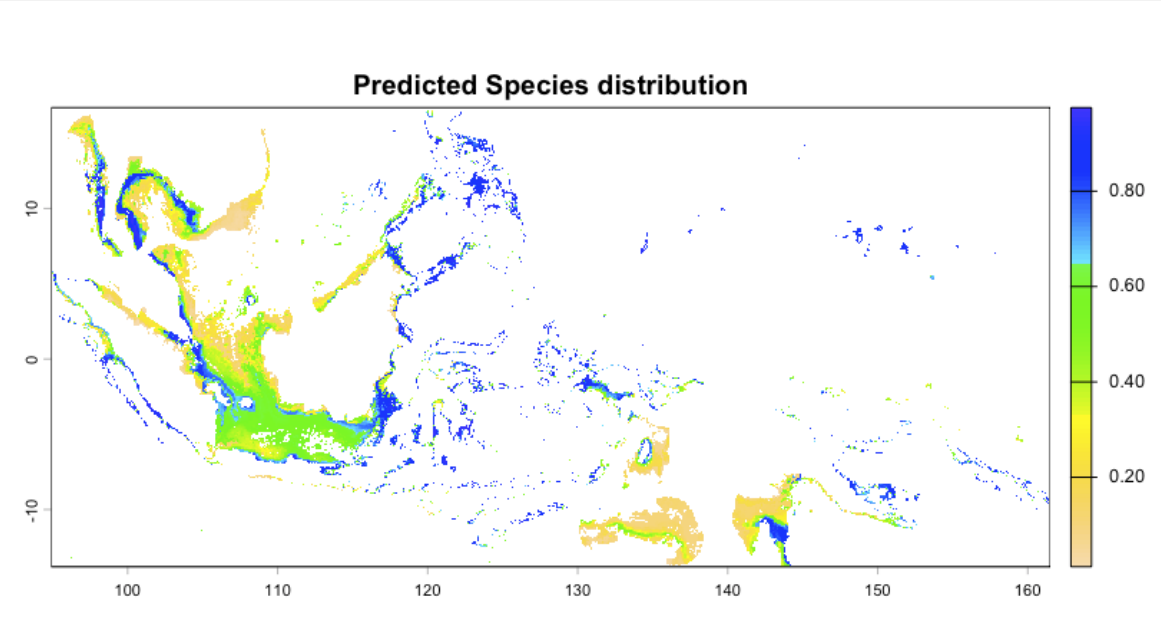
# inspect the data partitioning
modelData$plotDatasets
```

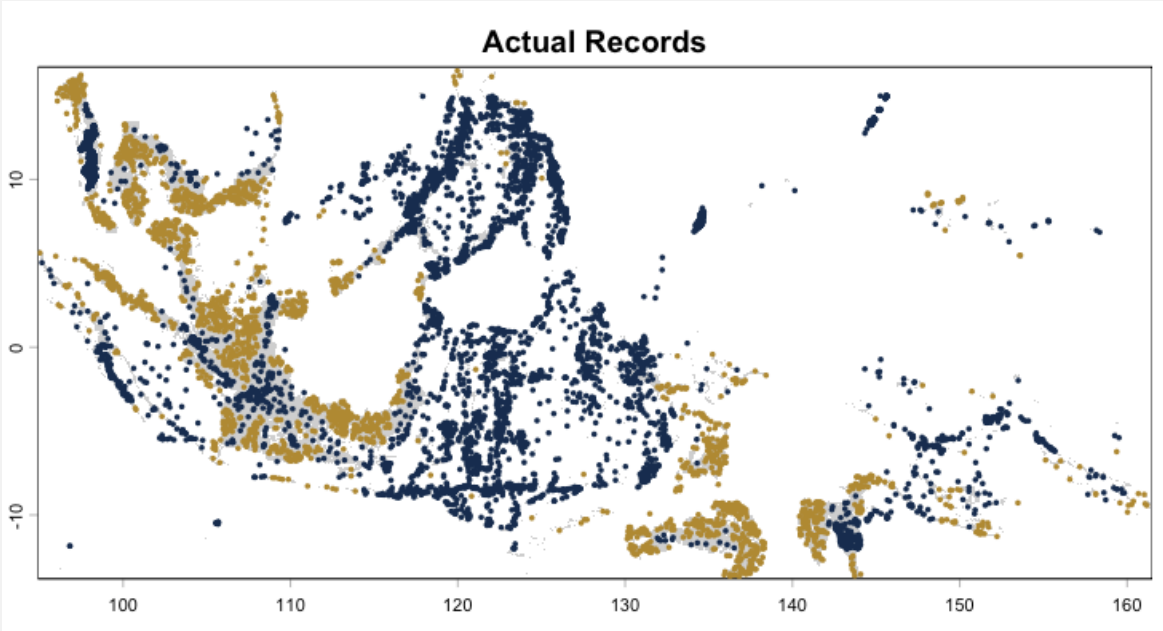


```r
# fit a BRT model to the dataset
model <- trainModel(modelData, algorithm = "brt")

# predict the distribution with the model over a raster stack
prediction <- predictModel(model = model, newData = environmentalLayers)

# map the predicted distribution
plot(prediction$rasterLayer, main = "Predicted Species distribution", col = rev(topo.colors(100)))
```





Challenge 1.1

The objective of this challenge is to **fit** a brt model to occurrence (presence / pseudo absence) and environmental data – Study region: Tropical Americas and Caribbean **(ext(-100, -55, 0, 32))**

Packages, data, and function to accomplish the challenge:

Files to use

"occurrenceRecords.csv"
"Present Benthic OceanTemperature LtMax.tif”"
"Present Benthic OceanTemperature LtMin.tif”"
"Present Benthic pH Mean.tif”"
“Present Benthic Salinity Mean.tif”
“Present Surface OceanTemperature LtMax.tif”
“Present Surface OceanTemperature LtMin.tif"

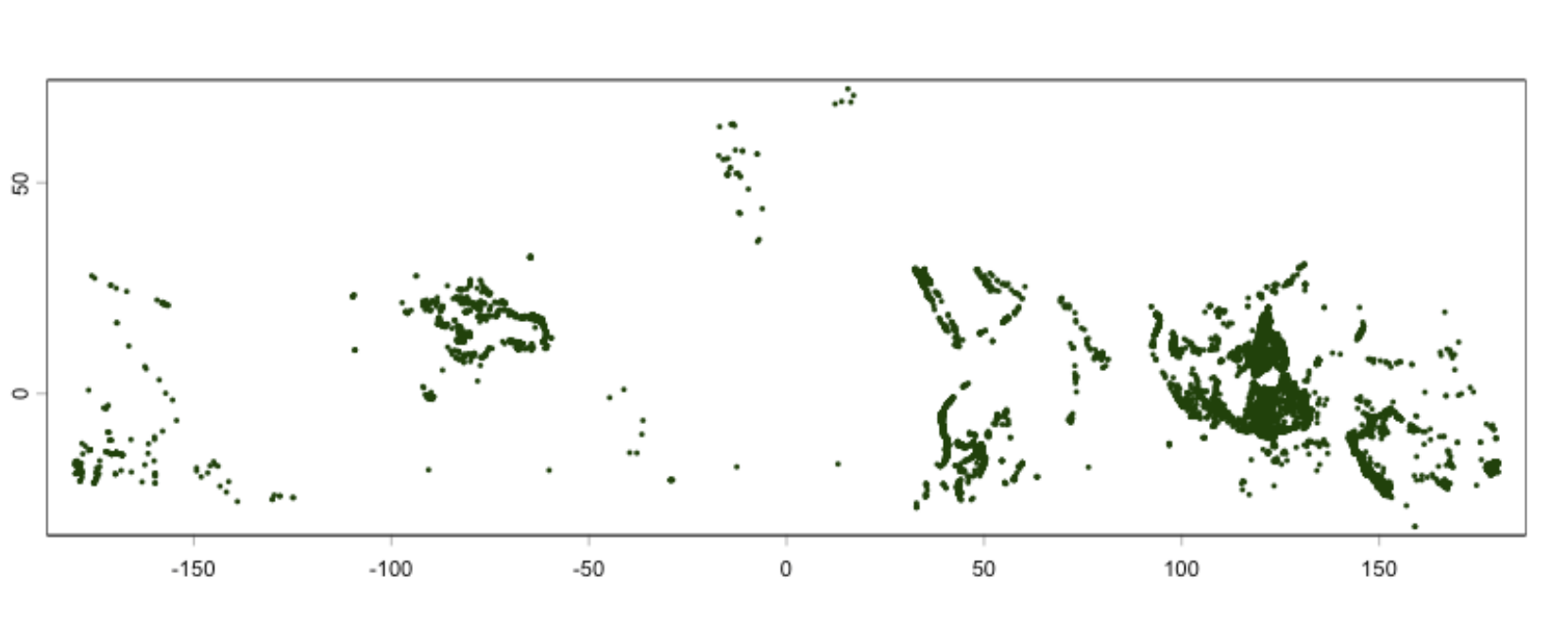
Functions to use

library (*if you are starting over again*)
rast
plot
studyRegion
pseudoAbsences
crop
mask
generateModelData
trainModel
predictModel

Presence / background data

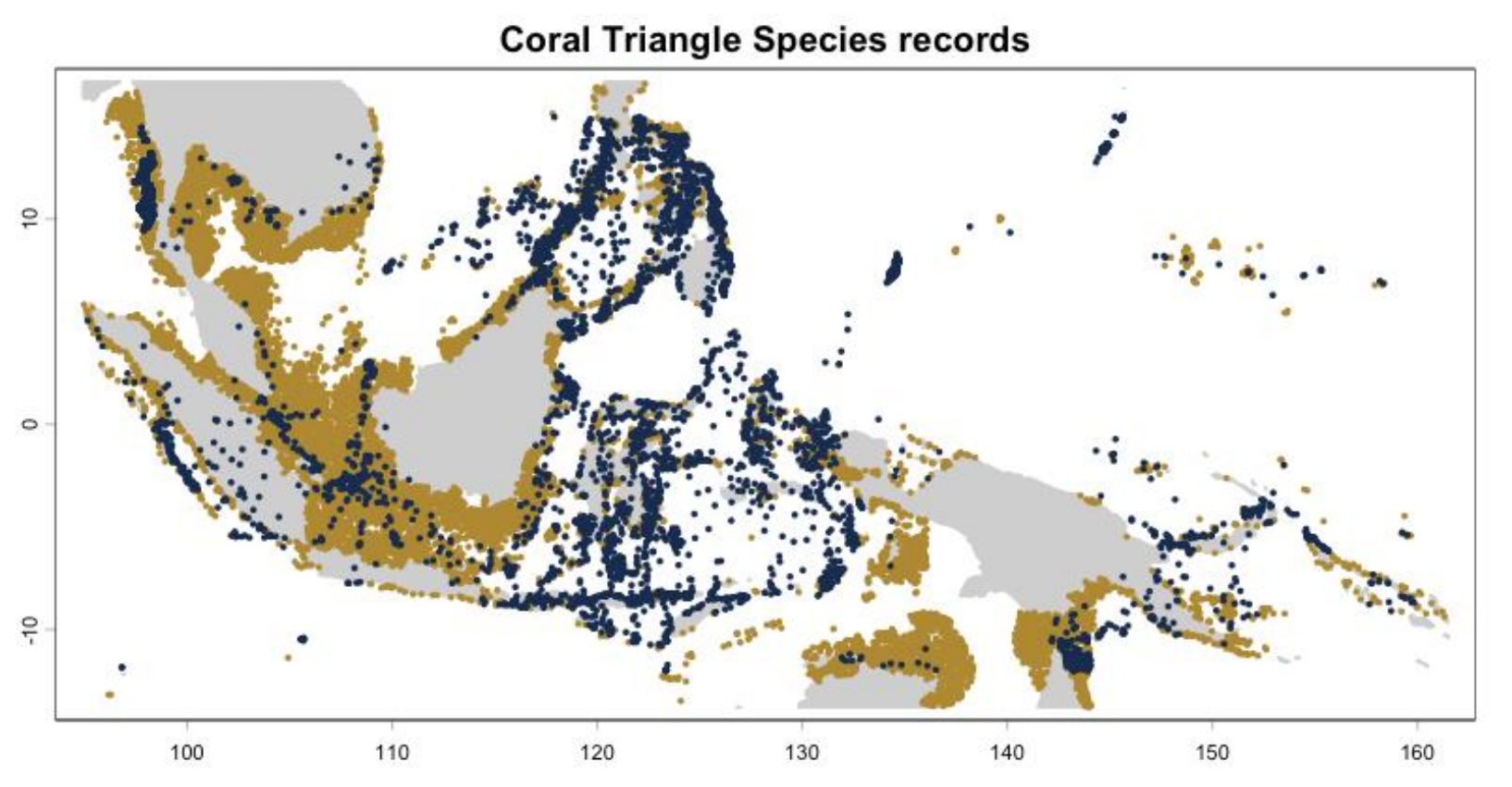
If **true absence data is not available**, but one wants to fit a maxent model, **background** data can be generated.

```
#Corals occurrence (Source: UNEP-WCMC)
presences <- read.csv("/Users/new/Desktop/Sci_Comp26/Data/Corals/occurrenceRecords.csv")
#transform data.frame into a spatial object
presences <- vect(presences, geom = c("Lon", "Lat"))
#plot occurrences
plot(presences, col = "#034B03", pch = 16, cex = 0.4) #pch = plotting character & cex = size
```

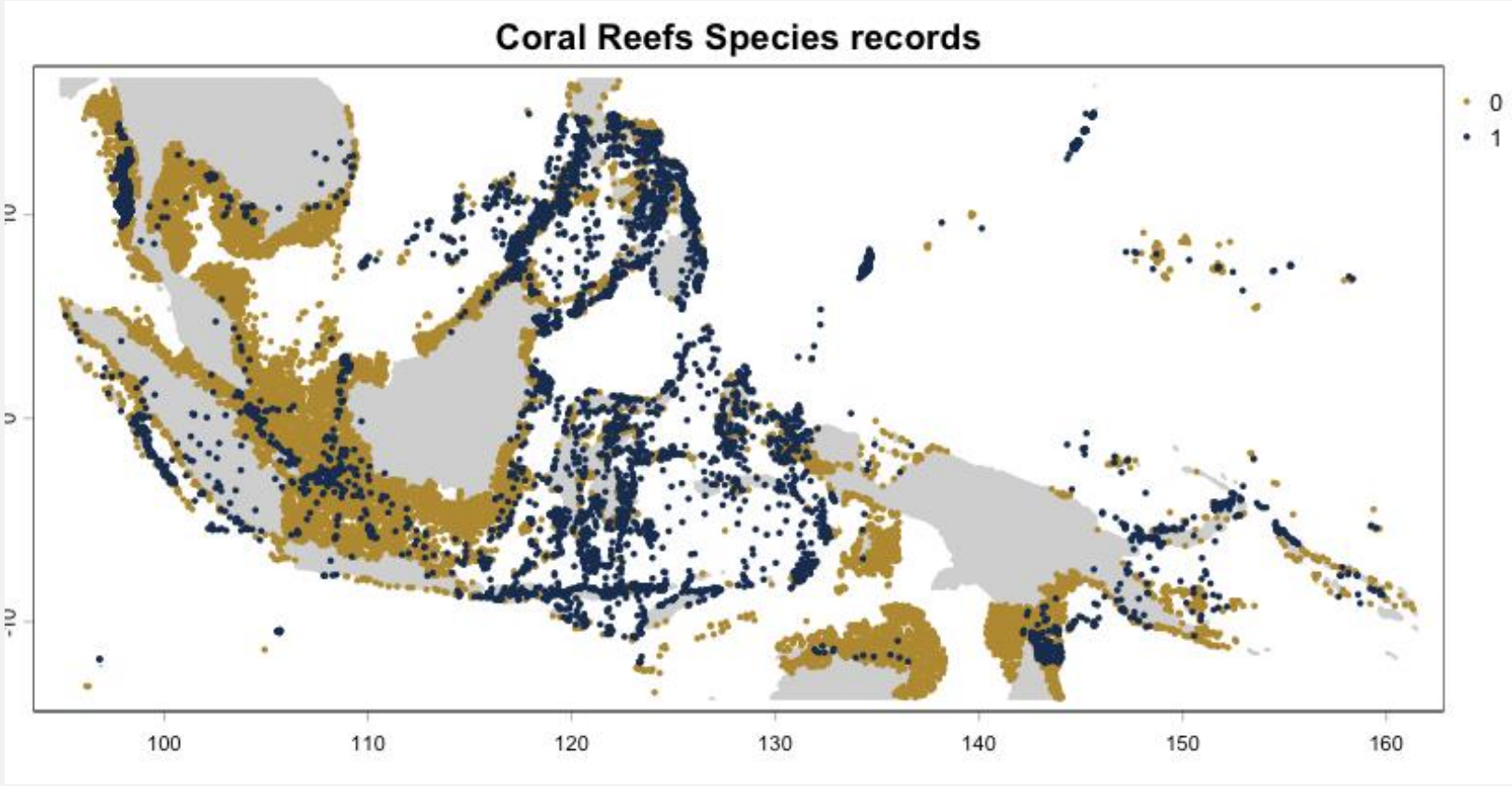


```r
#Let's do same but for another algorithm (MaxEnt)
# generate background points
background <- backgroundInformation(rasterLayers = environmentalLayers, n = 10000)

# plot records
plot(landmassCT, main = "Coral Triangle Species records", col = "#D4D4D4", border = "#D4D4D4", axes = TRUE)
plot(background, col = "#c29431", pch = 16, cex = 0.5, add = TRUE)
plot(presences_ct, col = "#043259", pch = 16, cex = 0.5, add = TRUE)
```



```r
# combine presences and background into a unique object for modelling
presences_ct$PA <- 1
background$PA <- 0
records <- rbind(background, presences_ct)
# plot records
plot(landmassCT, main = "Coral Reefs Species records", col = "#D4D4D4", border = "#D4D4D4", axes = TRUE)
plot(records, y = "PA", col = c("#c29431", "#043259"), pch = 16, cex = 0.5, add = TRUE)
```



```r
# extract environmental values and make an object with all information to model: Model evaluation with cross validation
modelData <- generateModelData(records, envConditions = environmentalLayers, method = "blocks", kFolds = 10)

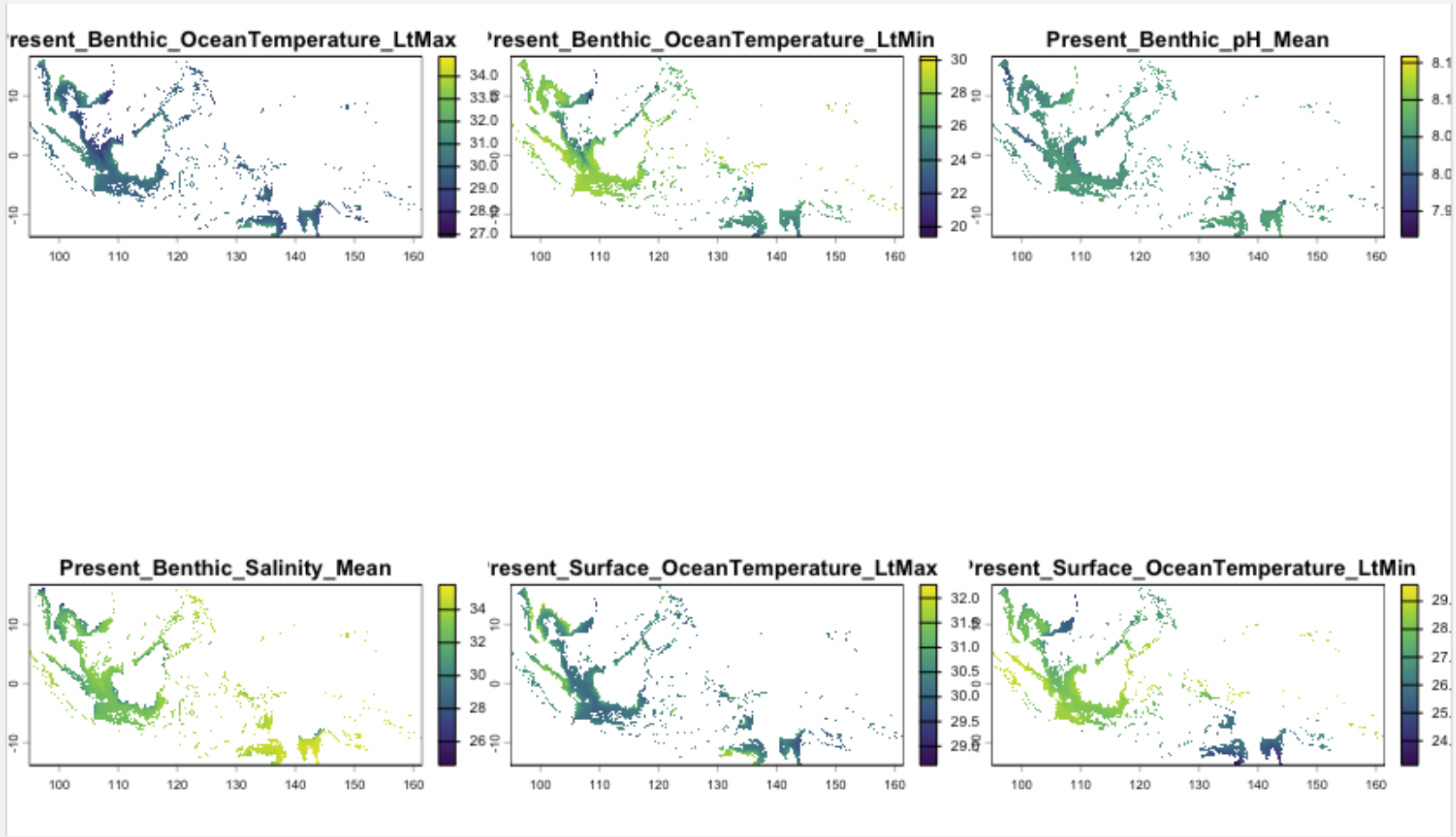
# inspect the data partitioning
modelData$plotDatasets
```



```r
# fit a BRT model to the dataset
model <- trainModel(modelData, algorithm = "maxent")
```

```
> model <- trainModel(modelData, algorithm = "maxent")
Error in trainModel(modelData, algorithm = "maxent") :
  In maxent algorithm, variable names cannot have spaces.
```

```r
#Rename layers for maxent to work
names(environmentalLayers) #Run this to check sequence of names for layers
names(environmentalLayers) <- c("Present_Benthic_OceanTemperature_LtMax",
                                "Present_Benthic_OceanTemperature_LtMin",
                                "Present_Benthic_pH_Mean",
                                "Present_Benthic_Salinity_Mean",
                                "Present_Surface_OceanTemperature_LtMax",
                                "Present_Surface_OceanTemperature_LtMin")
names(environmentalLayers) #Check if renaming worked
plot(environmentalLayers, axes = TRUE)
```



```r
# fit a BRT model to the dataset
model <- trainModel(modelData, algorithm = "maxent")
```

```r
# predict the distribution with the model over a raster stack
prediction <- predictModel(model = model, newData = environmentalLayers)


# map the predicted distribution
plot(prediction$rasterLayer, main = "Predicted Species distribution", col = rev(topo.colors(100)))
```

MaxEnt Takes a bit longer to run!!!!

Challenge 1.2

The objective of this challenge is to **fit** a maxent model to occurrence (presence / background) and environmental data – Study region: **Make your own choice**!

Packages, data, and function to accomplish the challenge:

Files to use

"occurrenceRecords.csv"
"Present Benthic OceanTemperature LtMax.tif”"
"Present Benthic OceanTemperature LtMin.tif”"
"Present Benthic pH Mean.tif”"
“Present Benthic Salinity Mean.tif”
“Present Surface OceanTemperature LtMax.tif”
“Present Surface OceanTemperature LtMin.tif"

Functions to use

library (*if you are starting over again*)
rast
plot
studyRegion
backgroundInformation
crop
mask
generateModelData
trainModel
predictModel